

Notes: Shleifer (JF, 1986)

Do Demand Curves for Stocks Slope Down?

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1 Big picture

- **Question.** Do stock demand curves slope down, or can investors buy and sell large amounts of equity without affecting price?
- **Setting.** The paper studies additions to the S&P 500 Index from 1966 to 1983. Inclusion triggers mechanical buying by index funds, so it provides a cleaner test than block-trade studies where buys or sells may also reveal private information.
- **Why S&P 500 additions are useful.** When a stock is added to the index, index funds buy it because they must track the benchmark, not because they discovered favorable news. This creates an outward shift in demand that is plausibly unrelated to fundamentals.
- **Core empirical challenge.** A positive price reaction could still reflect an information or certification effect if S&P inclusion tells the market that the firm is high quality or likely to survive for a long time.
- **Main baseline result.** Since September 1976, when S&P started an early warning service that makes announcement timing precise, the average abnormal return on the announcement date is about 2.79%, with a t -statistic of 12.4.
- **Persistence result.** The announcement gain does not reverse quickly. Most of it remains for at least 10 to 20 trading days.
- **Quantity result.** The price effect is larger when abnormal announcement-day volume is larger, which is what one would expect if buying by index funds pushes prices up.
- **Time-series result.** The abnormal return is larger in 1981-1983 than in 1976-1980, consistent with the rapid growth of index funds over time.
- **Main anti-certification result.** The inclusion effect is not related to S&P bond ratings, which weakens the idea that S&P has special information about firms' longevities and reveals it through index inclusion.
- **Bottom-line conclusion.** The evidence is most consistent with downward-sloping demand curves for individual stocks. The paper does not rule out every alternative story, but it makes the pure horizontal-demand view very hard to sustain.
- **Economic takeaway.** If stock demand curves slope down, then supply shifts can move prices even without new information. That matters for how one thinks about corporate finance, indexing, and asset-pricing models built on nearly horizontal demand curves.

2 Data and measurement (key objects)

2.1 Why S&P 500 additions are a useful laboratory

- Each year S&P removes firms from the index, usually because of mergers, acquisitions, or takeovers, and simultaneously adds replacements.

- Neubert's description of the selection rule lists six criteria for inclusion: size, industry classification, capitalization, trading volume/turnover, emerging companies or industries, and responsiveness of the stock price to industry affairs.
- These criteria are public and, as the paper emphasizes, none is directly about the firm's future performance.
- After inclusion, index funds and other self-indexing investors buy the stock. In recent years of the sample, this could amount to as much as roughly 3% of the firm's equity.

Interpretation. The test works because inclusion moves demand for reasons that are largely mechanical. If demand curves are horizontal, that buying should not move price. If they slope down, price should jump.

2.2 Announcement dates and the key pre/post-1976 split

The event-study design depends on dating the first moment at which the market learns that the stock will be added.

- **After September 1976.** S&P operated an early notification service. Changes were made effective after the market close, and subscribers were informed within minutes. The next trading day is therefore the announcement date, denoted *AD*.
- **Before September 1976.** Changes were recorded in the monthly *Cumulative Index to Standard and Poor's Outlook* (CISPO). The paper assumes that the announcement date is the day subscribers received the CISPO issue containing the change.

Interpretation. The post-1976 subsample is the cleaner design because the announcement date is measured much more precisely. The paper therefore presents the two periods separately.

2.3 Sample construction

The starting sample consists of 331 firms added to the S&P 500 between 1966 and 1983.

The paper removes:

- 34 firms with no CRSP data,
- 13 perfectly anticipated inclusions,
- 17 firms involved in the June 30, 1976 major revision of the index,
- and 21 early-period firms for which announcement dates could not be ascertained reliably.

This leaves a final sample of 246 firms, or about 74% of the initial sample.

The main split is:

- 144 firms in 1966-1975,
- 102 firms in September 1976-1983.

Interpretation. The empirical results are not driven by a hand-picked list of examples. They come from a broad sample of index additions over nearly two decades.

2.4 Event-study objects

The paper uses the standard daily market-model event-study framework, estimated with the CRSP value-weighted market portfolio. To allow for the possibility that inclusion changes a stock's systematic risk, return equations are estimated separately before and after the inclusion.

The basic return model is

$$R_{i\tau} = \alpha_i + \beta_i R_{m\tau} + \varepsilon_{i\tau}, \quad (1)$$

where τ indexes event time relative to the announcement date.

For each firm and event day, the prediction error is

$$PE_{i\tau} = R_{i\tau} - \hat{\alpha}_i - \hat{\beta}_i R_{m\tau}. \quad (2)$$

The average prediction error is

$$APE_\tau = \frac{1}{N} \sum_{i=1}^N PE_{i\tau}, \quad (3)$$

and the cumulative average prediction error from τ_1 to τ_2 is

$$CAPE_{\tau_1, \tau_2} = \sum_{\tau=\tau_1}^{\tau_2} APE_\tau. \quad (4)$$

Interpretation.

- APE_τ measures abnormal performance on one event day.
- $CAPE$ measures the total abnormal performance over a window.
- If the price increase at inclusion quickly reverses, that looks like temporary price pressure. If it persists, the evidence is more consistent with a downward-sloping demand curve.

2.5 Measures of index-fund buying

The paper uses volume to proxy for the size of the demand shift caused by indexing.

- **Abnormal announcement-day volume (ABADVOL).** Announcement-day trading volume minus the stock's average daily volume over the previous six months, expressed as a fraction of shares outstanding.
- **Announcement-day volume (ADVOL).** Raw trading volume on the announcement date.
- **Usual volume (USVOL).** Average normal volume for the stock, included to capture how easy it is to absorb buying in a more liquid name.
- **Abnormal announcement-week volume.** A noisier measure meant to capture buying spread over several days rather than concentrated at once.

For the post-1976 subsample with volume data, the average abnormal announcement-day volume is 0.340% of shares outstanding, and the average abnormal announcement-week volume is 1.012%.

Interpretation. These are noisy measures, but if prices move because indexers must buy the stock, larger abnormal buying should be associated with larger abnormal returns.

2.6 Variables for the competing information and liquidity stories

To test alternative explanations, the paper constructs several additional objects.

- **Bond-rating dummies.**

- $DUM1 = 1$ if the firm's S&P bond rating is A or better.
- $DUM2 = 1$ if the firm has no S&P bond rating.
- The omitted category is firms with ratings below A .

In the 84-observation post-1976 volume sample, 26 firms are in the high-rating group, 48 have no rating, and 10 are below A .

- **Fortune 500 membership.** Used as a rough proxy for whether the firm was already large, visible, and closely followed before S&P inclusion.

Interpretation.

- If S&P inclusion is mainly certification, firms with weaker prior public certification should react more.
- If the mechanism is mainly liquidity, lesser-known firms should react more because they gain more visibility and trading interest.

2.7 Key descriptive facts

The main sample facts that organize the paper are:

- No meaningful announcement-date effect before September 1976.
- A very large positive announcement-date effect after September 1976.
- No clear price run-up before the announcement date.
- No statistically significant reversal in the 20 trading days after the announcement.

Interpretation. The descriptive picture already leans toward the demand-curve story: a sharp, unanticipated jump when the index inclusion becomes public, with little evidence that the market had already priced it in.

3 Models and empirical specifications (all equations + interpretation)

3.1 The main economic mechanism

The paper's core null and alternative are simple.

- **Horizontal-demand null.** If close substitutes make stock demand essentially flat, the forced buying by index funds should have little or no price effect.

- **Downward-sloping-demand (DS) hypothesis.** If the stock's demand curve slopes down, forced buying shifts demand outward and raises price.

A competing interpretation is the **information or certification hypothesis**: inclusion in the S&P 500 may tell the market that S&P regards the company as high quality or unusually durable.

Interpretation. The whole paper is a comparison between these stories using timing, persistence, quantities, and cross-sectional patterns.

3.2 Baseline event-study specification

The empirical backbone of the paper is the market-model event study described above. The main objects reported are:

- $CAPE_{AD-20,AD-1}$: pre-announcement abnormal return,
- APE_{AD} : abnormal return on the announcement date,
- $CAPE_{AD+1,AD+10}$ and $CAPE_{AD+11,AD+20}$: post-announcement drift or reversal,
- $CAPE_{AD,AD+k}$: cumulative abnormal return starting on the announcement date.

Interpretation.

- A large APE_{AD} is the immediate price-impact estimate.
- A $CAPE$ beginning at AD tells us whether the gain survives.
- The paper is especially interested in whether prices fall back after inclusion, because quick reversal would weaken the case for a durable demand-curve shift.

3.3 Time variation with the growth of index funds

A central implication of the DS hypothesis is that the price response should be larger when the quantity of forced buying is larger.

The paper documents that index funds owned less than 0.5% of the S&P 500 in 1975, about 1.4% in 1979, and about 3.1% in 1983. It therefore splits the post-September-1976 sample into:

- September 1976-1980,
- 1981-1983.

Interpretation. If the effect grows alongside indexing, that is naturally consistent with a quantity-based demand story.

3.4 Cross-sectional quantity regressions

To see whether larger buying pressure is associated with larger abnormal returns, the paper estimates two post-1976 regressions.

Regression (1): abnormal return on abnormal volume

$$RETURN = 2.41 + 1.22 ABADVOL, \quad \bar{R}^2 = 0.04, \quad N = 84. \quad (5)$$

The reported t -statistics are (7.55) on the constant and (2.13) on $ABADVOL$.

Interpretation. The positive slope says that stocks with more abnormal announcement-day buying have larger abnormal announcement-day price jumps.

Regression (2): abnormal return on raw and usual volume

$$RETURN = 2.77 + 0.982 ADVOL - 1.59 USVOL, \quad \bar{R}^2 = 0.05, N = 84. \quad (6)$$

The reported t -statistics are (6.73), (1.64), and (−2.51), respectively.

Interpretation.

- The positive coefficient on $ADVOL$ means that more buying is associated with higher announcement returns.
- The negative coefficient on $USVOL$ means that more liquid, more heavily traded stocks have smaller price effects.
- Together, these are exactly the patterns one would expect if forced demand moves price more when it is harder to absorb.

3.5 Test of the certification hypothesis using bond ratings

The paper asks whether inclusion matters more for firms whose S&P bond ratings previously looked weak or absent. The estimated regression is

$$RETURN = 2.34 + 0.356 DUM1 + 0.660 DUM2, \quad \bar{R}^2 = -0.02, N = 84. \quad (7)$$

The reported t -statistics are (3.07), (0.397), and (0.837).

Interpretation. There is no statistically meaningful relation between the inclusion effect and bond-rating categories. That weakens the view that S&P is using inclusion to reveal special information about firm longevity.

3.6 Permanence tests

The paper uses cumulative abnormal returns from AD forward to ask whether the announcement gain disappears quickly:

$$CAPE_{AD, AD+k}, \quad k \in \{5, 10, 20, 60\}. \quad (8)$$

Interpretation. These tests do double duty. They ask both whether prices fall after the announcement and whether any such decline is large enough to wipe out the initial gain.

3.7 Other explanations considered in the paper

The paper discusses three alternatives in detail.

- **Transaction costs.** Buyer-initiated trades may require sellers to be compensated for rebalancing costs.

- **Market segmentation.** Some investors only want S&P 500 stocks, so inclusion expands the buyer base.
- **Liquidity.** Inclusion may make the stock better known, more widely followed, and more liquid, lowering required returns.

Interpretation. The paper's view is that market segmentation is really a special case of downward-sloping demand curves, while transaction costs and liquidity are independent competing explanations that must be bounded or tested.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the main effect

There is no instrumental variable in the modern sense. Identification instead comes from the event-study logic:

- S&P index additions create mechanical buying pressure from indexers.
- The announcement date is sharp after September 1976.
- The paper studies abnormal returns relative to a market model and focuses on a narrow event window.

Interpretation. The causal question is: when a stock is unexpectedly announced as a new S&P 500 constituent, what happens to price?

4.2 Why reverse causality is weak

The usual reverse-causality concern in event studies would be that price movements cause the event. Here that story is hard to sustain in the short event window because:

- the event is a public announcement by S&P,
- index-fund buying follows from inclusion rather than preceding it,
- and the paper finds no evidence of a price run-up right before the announcement date.

Interpretation. The results do not look like “prices moved first and therefore S&P reacted.” They look like prices jumped when the index addition became public.

4.3 Why the announcement date is credible

The most important institutional detail is the September 1976 early warning service. After that point:

- S&P announces changes within minutes after the market closes,

- and the next trading day cleanly captures the market reaction.

For the earlier period, the paper uses CISPO mailing and publication dates, which are noisier. The absence of pre-announcement abnormal returns in that earlier period helps reassure the reader that announcement-date mismeasurement is not driving the post-1976 result.

Interpretation. The clean timing is one reason the paper's strongest evidence comes from the post-1976 subsample.

4.4 Why the information explanation looks incomplete

The paper does not claim that information effects are logically impossible. It argues they are implausible as the full story for four reasons.

1. S&P's stated inclusion criteria are public and not obviously about future profits.
2. There is no evidence that the market anticipates the inclusion before the announcement.
3. The abnormal return grows as index funds become larger.
4. The return is unrelated to bond-rating categories that should matter if S&P were certifying firm quality or longevity.

Interpretation. A pure certification story has trouble fitting the timing, quantity, and cross-sectional evidence all at once.

4.5 What the persistence result contributes

The price effect does not vanish immediately after the announcement. The point estimates show only small declines after AD , and those declines are not statistically significant.

Interpretation. If the event were only a temporary burst of uninformed order flow that the market could instantly absorb, one might expect sharp reversal. The paper instead finds that most of the gain remains for at least 10 to 20 trading days.

4.6 Why the quantity evidence matters

The regressions using abnormal volume are noisy, but they move the paper beyond a pure event-study correlation.

- Larger abnormal buying is associated with larger announcement-date returns.
- Stocks with higher usual volume have smaller price effects.

Interpretation. This is exactly the comparative-static prediction of a downward-sloping demand curve: a bigger demand shock should move price more, especially where the market is less able to absorb it.

4.7 Why transaction costs do not seem large enough

Shleifer argues that by the early 1980s institutional trading costs had fallen to roughly 50 basis points per share, with only part of that reflecting price impact rather than brokerage or bid-ask costs.

If sellers to index funds are also institutions, then under horizontal demand curves they should require at most about 1% compensation to sell. The observed price effects in the early 1980s exceed 3%.

Interpretation. Transaction costs may explain some fraction of the premium, but they look too small to account for the whole effect.

4.8 Liquidity and market-segmentation arguments

Market segmentation. Investors who only want S&P 500 stocks may buy newly included names. The paper's response is that this still requires the supply faced by initial holders to slope down. So market segmentation is treated as a special case of the DS hypothesis rather than a true rival.

Liquidity. If inclusion improves visibility and lowers the bid-ask spread, lesser-known firms should react more. The paper compares Fortune 500 and non-Fortune 500 stocks and finds no difference in excess returns.

Interpretation. The liquidity story is not supported by the paper's simple cross-sectional test.

4.9 What the paper can and cannot distinguish

The paper is strongest in rejecting the strictly horizontal-demand view. It is weaker in separately identifying every channel behind the price reaction.

- It does not fully separate downward-sloping demand from every possible segmentation story.
- It cannot say exactly how long the price effect lasts beyond the first few weeks, because standard errors grow quickly over longer windows.
- It does not estimate a structural elasticity of demand; it documents event-study price impact and supporting comparative statics.

Interpretation. This is an early and persuasive reduced-form paper, not a fully structural demand estimate.

5 Tables and figures

5.1 Table I: average abnormal returns surrounding inclusion

Read:

- Before September 1976, the average announcement-date abnormal return is -0.192% , statistically insignificant.

- After September 1976, the announcement-date abnormal return is 2.79% with a t -statistic of 12.4.
- In the 20 trading days before the announcement, cumulative abnormal returns are slightly negative rather than positive in both subsamples.
- In the 10 trading days after the announcement, the point estimate is -0.859% in the post-1976 sample, but this is not statistically significant.

Table I
**Average Abnormal Returns Surrounding Inclusion of
Stocks into the S&P 500 Index**

Days relative to the Announcement Date (AD)	Average Cumulative Prediction Error	
	1966–1975 (before the early warning service)	Sept, 1976–1983 (after the early warning service)
	N = 144	N = 102
AD – 20 through AD – 1	–2.86 (–2.85)	–1.49 (–1.25)
AD	–0.192 (–0.918)	2.79 (12.4)
AD + 1 through AD + 10	–0.065 (–0.091)	–0.859 (–1.03)
AD + 11 through AD + 20	1.12 (1.57)	–0.154 (–0.184)

Notes:

1. t -statistics are included in parentheses
2. details of calculations are provided in the text

Economic message. The big fact is the sharp post-1976 announcement-day jump. The lack of pre-run-up says the event is not anticipated, and the lack of a statistically meaningful reversal says the gain is not immediately unwound.

5.2 Table II: persistence of abnormal returns and growth over time

Read:

- In the full post-1976 sample, the cumulative abnormal return from AD through $AD + 5$ is 2.22% with $t = 3.46$.
- From AD through $AD + 10$, it is still 1.94% with $t = 2.20$.
- From AD through $AD + 20$, the point estimate remains 1.78%, and even though the standard error is larger, the gain is still about one percentage point below the original announcement effect.

- The announcement-day effect is 2.27% in 1976-1980 and 3.19% in 1981-1983. The paper reports that a one-sided equality test rejects at roughly the 98% confidence level.

Table II
Tests of Persistence of Abnormal Returns Subsequent to
Inclusion into S&P 500

Days over which the abnormal return is cumulated	Average Cumulative Prediction Error		
	Full Sample Sept, 1976–1983	Early Subsample Sept, 1976–1980	Later Subsample 1981–1983
AD	2.79 (12.4)	2.27 (7.63)	3.19 (10.1)
AD through AD + 5	2.22 (3.46)	1.58 (1.88)	2.70 (3.10)
AD through AD + 10	1.94 (2.20)	1.09 (0.946)	2.58 (2.17)
AD through AD + 20	1.78 (1.46)	−0.807 (−0.505)	3.77 (2.28)
AD through AD + 60	1.71 (0.819)	−1.05 (−0.383)	3.85 (1.36)

Notes:

1. *t*-statistics in parentheses
2. details of calculation of abnormal returns are provided in the text

Economic message. This table does two important jobs. First, it shows that most of the inclusion gain persists for at least 10 to 20 days. Second, it shows that the effect gets larger as index funds get larger, which is exactly what the DS hypothesis predicts.

5.3 Quantity evidence: equations (1) and (2)

Read:

- The average abnormal announcement-day volume after September 1976 is 0.340% of shares outstanding; abnormal announcement-week volume is 1.012%.
- In regression (1), the coefficient on abnormal announcement-day volume is 1.22 with $t = 2.13$.
- In regression (2), the coefficient on raw announcement-day volume is 0.982 with $t = 1.64$, while the coefficient on usual volume is -1.59 with $t = -2.51$.

Economic message. Bigger buying pressure goes with bigger price effects, and liquid stocks absorb the same buying more easily. The volume evidence is noisy, but it pushes the paper beyond a pure event-timing result.

5.4 Information and alternative explanations

Read:

- In regression (3), neither the high-bond-rating dummy nor the no-rating dummy is significantly related to the abnormal announcement return.
- Transaction costs appear too small to explain a 3%-plus price effect in the early 1980s.
- The market-segmentation story is treated as a special case of downward-sloping demand rather than a true rival.
- The liquidity hypothesis predicts higher abnormal returns for lesser-known firms, but the paper finds no Fortune 500 versus non-Fortune 500 difference.

Economic message. The paper does not reject every imaginable alternative, but it steadily narrows the set of plausible ones. The main remaining interpretation is that stock demand curves are not flat.

6 Short summary

- The paper uses S&P 500 additions as a clean demand-shift event to test whether stock demand curves slope down.
- The event-study evidence is weak before September 1976, but after the early warning service begins, newly included stocks earn an average announcement-date abnormal return of about 2.79%.
- Most of that gain remains for at least 10 to 20 trading days.
- The effect is larger in later years, when index funds are bigger.
- Announcement-day abnormal returns are positively related to abnormal trading volume and negatively related to usual volume.
- The price effect is not related to S&P bond ratings, which weakens the certification story.
- Transaction costs and liquidity do not seem large enough to explain the whole effect.
- The broad conclusion is that demand curves for individual stocks slope down.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that when a stock is newly added to the S&P 500, its price rises sharply on the announcement and does not immediately fall back. The size, timing, persistence, and quantity patterns are all consistent with downward-sloping demand curves for stocks.

7.2 Economic intuition

The mechanism is simple. Index funds must buy the stock because benchmark composition changed. If the stock had perfect substitutes and a horizontal demand curve, that buying would not materially

affect price. Instead, the price moves by several percent, which implies that the market cannot absorb the shift at a constant price.

7.3 Takeaway

Shleifer's message is broader than the S&P 500 event itself. Stock prices are not pinned down only by fundamentals plus frictionless arbitrage. Changes in demand can matter. That insight has implications for indexing, corporate finance, and any asset-pricing argument that relies on nearly horizontal demand for individual securities.